MODULE - 4

(Infosys Springboard Virtual Internship 6.0)

SUBMITTED BY - ASHISH CHAUHAN

Table Of Content:

1. Objective
2. Dataset Used
3. Measures Created
4. Visualizations
5. Key Insights
6. Conclusion

1. Objective

The objective of this module is to analyze hotel booking patterns and identify operational and financial inefficiencies using forecasting and trend analysis. The study focuses on understanding booking demand, cancellation behavior, lead time distribution, and the impact of refunds and no-shows.

This analysis helps in:

- Predicting future booking demand
- Identifying high-risk periods of cancellations and no-shows
- Reducing revenue loss through better planning
- Improving customer management strategies

2. Dataset Used

The analysis is performed using the Bookings dataset, which contains transactional and behavioral data related to hotel reservations.

Key fields include:

- BookingID – Unique identifier for each booking
- MonthStart – Time dimension for trend analysis
- LeadTime (days) – Number of days between booking and check-in
- CancellationFlag – Indicates whether booking was cancelled
- RefundAmount – Amount refunded due to cancellation
- NoShowFlag – Indicates if customer did not show up
- CustomerRevenue – Revenue generated per booking

This dataset enables both time-series analysis and behavioral insights.

onthStart	GuestType	NumOfBookings	CustomerRevenue	CustomerCluster	CancellationFlag	RefundAmount	RefundFlag	NoShowFlag	RoomRate	LeadTime(days) (bins)
Vovember 2024	Solo	12	622973	High Value	No	0	1	0	1800	20
Vovember 2024	Business	16	1211968	High Value	No	0	1	0		
February 2024	Solo	17	990583	High Value	No	0	1	0		
1 January 2025	Solo	27	2166707	High Value	No	0	1	0		
eptember 2024	Solo	17	1000900	High Value	No	0	1	0	6800	0
1 January 2026	Solo	12	658090	High Value	No	0	1	0	6000	0
1 January 2025	Corporate	21	880445	High Value	No	0	1	0	14000	70
Vovember 2025	Corporate	14	907161	High Value	No	0	1	0	10000	50
01 June 2025	Solo	14	722654	High Value	No	0	1	0	3500	0
Vovember 2025	Corporate	15	591073	High Value	No	0	1	0	3200	80
1 October 2025	Solo	19	1136388	High Value	No	0	1	0	6000	90
Vovember 2024	Corporate	16	762824	High Value	No	0	1	0	3200	40
1 January 2025	Solo	12	838656	High Value	No	0	1	0	13000	50
Vovember 2025	Solo	15	1304624	High Value	No	0	1	0	18000	30
01 July 2025	Corporate	15	1006366	High Value	No	0	1	0	13000	20
eptember 2025	Solo	16	1125704	High Value	No	0	1	0	14000	20
1 October 2025	Solo	15	1380041	High Value	No	0	1	0	13000	80
1 October 2025	Solo	18	1102150	High Value	No	0	1	0	8000	50
eptember 2024	Solo	6	178097	Regular	No	0	1	0	3200	70
December 2025	Solo	18	1415378	High Value	No	0	1	0	7500	70
01 March 2024	Solo	15	607527	High Value	No	0	1	0	7000	0
February 2026	Solo	11	666878	High Value	No	0	1	0	1800	20
01 March 2025	Solo	15	992285	High Value	No	0	1	0	4400	0
1 January 2024	Solo	9	554858	High Value	No	0	1	0	6800	70
December 2025	Solo	8	441020	Regular	No	0	1	0	5200	30
Vovember 2025	Solo	10	229824	Regular	No	0	1	0	14000	70

Name Bookings
Storage mode Import
Data refreshed 26/3/2026, 6:49:12 pm

Bookings (25,000 rows)

3. Measures Created

- **NoShowCount**
  Counts the number of bookings where customers did not show up.

  NoShowCount = CALCULATE(COUNTROWS(Bookings), Bookings[BookingStatus] = "No-Show")

- **NoShowRate**
  Represents the percentage of no-shows out of total bookings.

  NoShowRate = DIVIDE([NoShowCount], COUNTROWS(Bookings))

- **High NoShowRate = poor customer commitment**
- **Leads to empty rooms and lost revenue**

4. Visualizations

Visual 1: Booking Demand Forecast

This visual shows historical booking trends along with forecasted demand using time-series forecasting.

Insights:

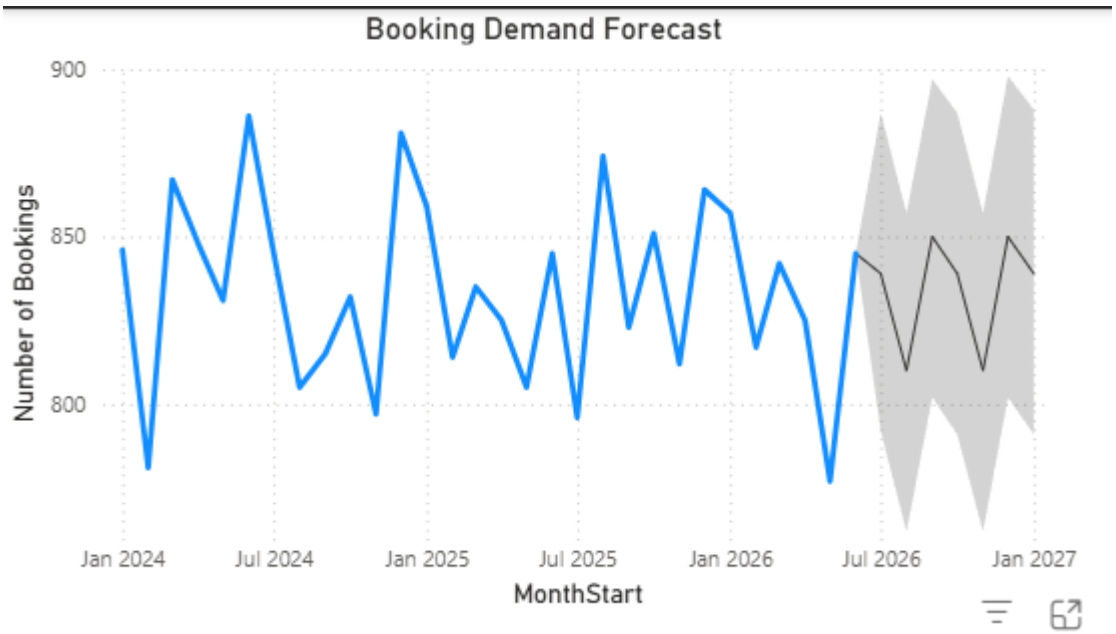
- Booking demand remains relatively stable with periodic fluctuations
- Forecast indicates consistent future demand
- No extreme spikes or drops → stable business

Problem Identified:

- Lack of strong growth trend may indicate stagnation

Solution:

- Introduce promotional campaigns during low-demand periods
- Use dynamic pricing strategies



Visual 2: Cancellation Rate Trend

Shows variation in cancellation rates across different months.

Insights:

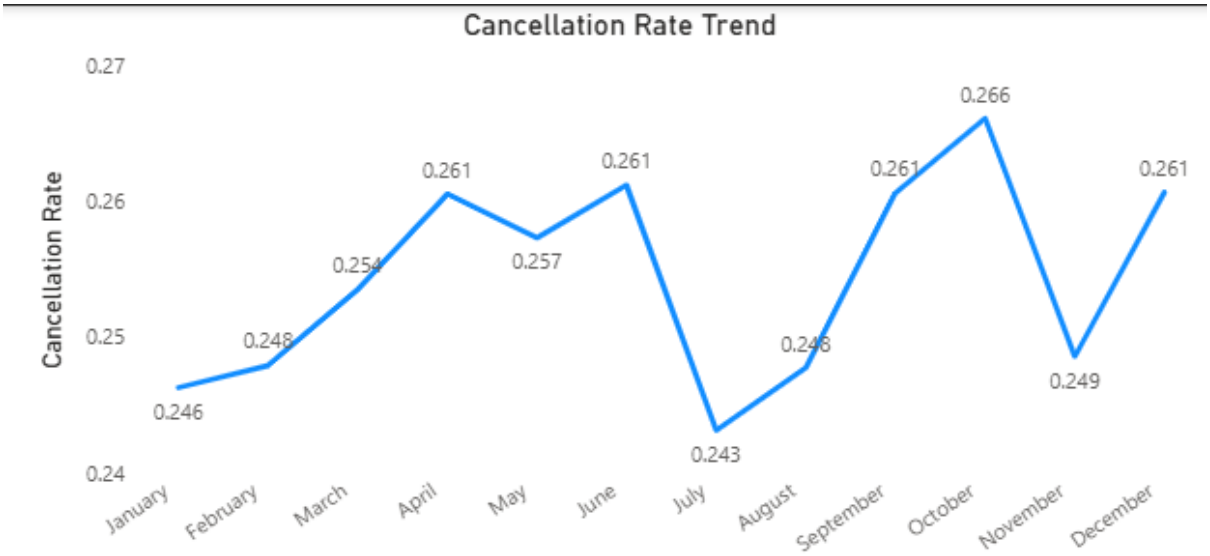
- Certain months have higher cancellation rates
- Indicates seasonal uncertainty or pricing mismatch

Problem Identified:

- High cancellations lead to revenue unpredictability

Solution:

- Implement stricter cancellation policies
- Offer incentives for non-refundable bookings
- Send reminders before check-in



Visual 3: Booking Distribution by Lead Time

Displays how far in advance customers book rooms.

Insights:

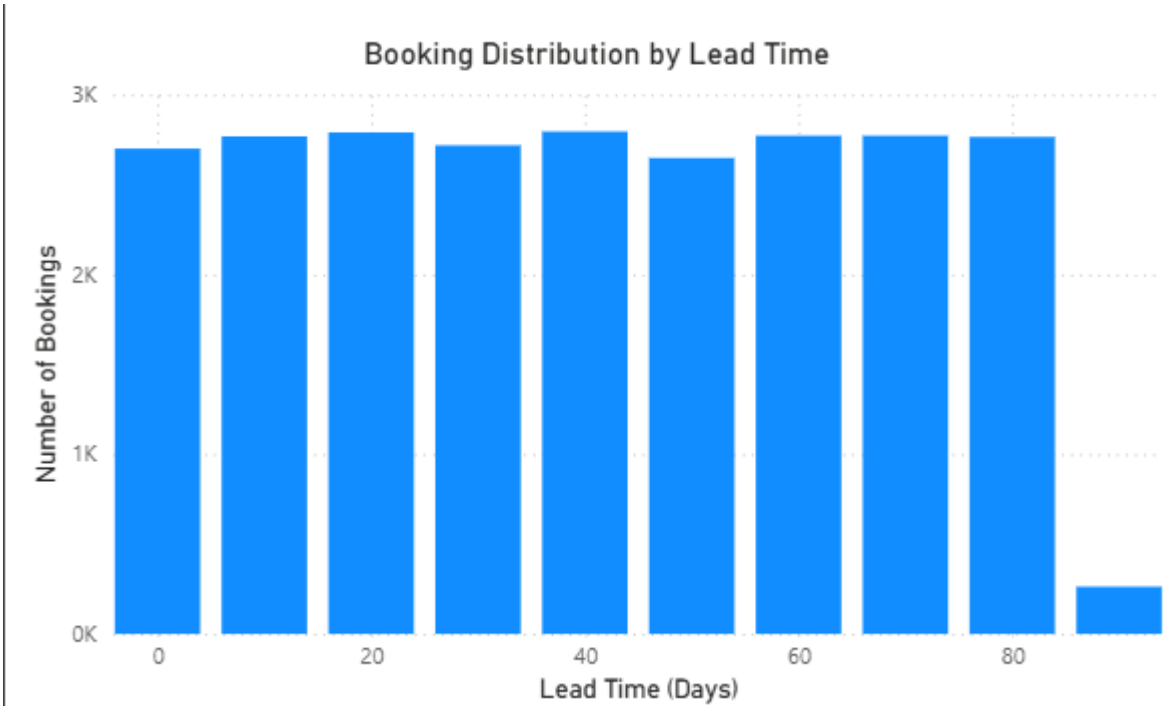
- Majority bookings occur in mid-range lead times
- Very early and last-minute bookings are limited

Problem Identified:

- Low early bookings reduce planning efficiency
- Last-minute bookings create operational pressure

Solution:

- Offer discounts for early bookings
- Use surge pricing for last-minute bookings



Visual 4: Monthly Refund vs No-Show Trends

A combo chart comparing:

- Refund Amount (financial loss)
- No-Show Rate (operational inefficiency)

Insights:

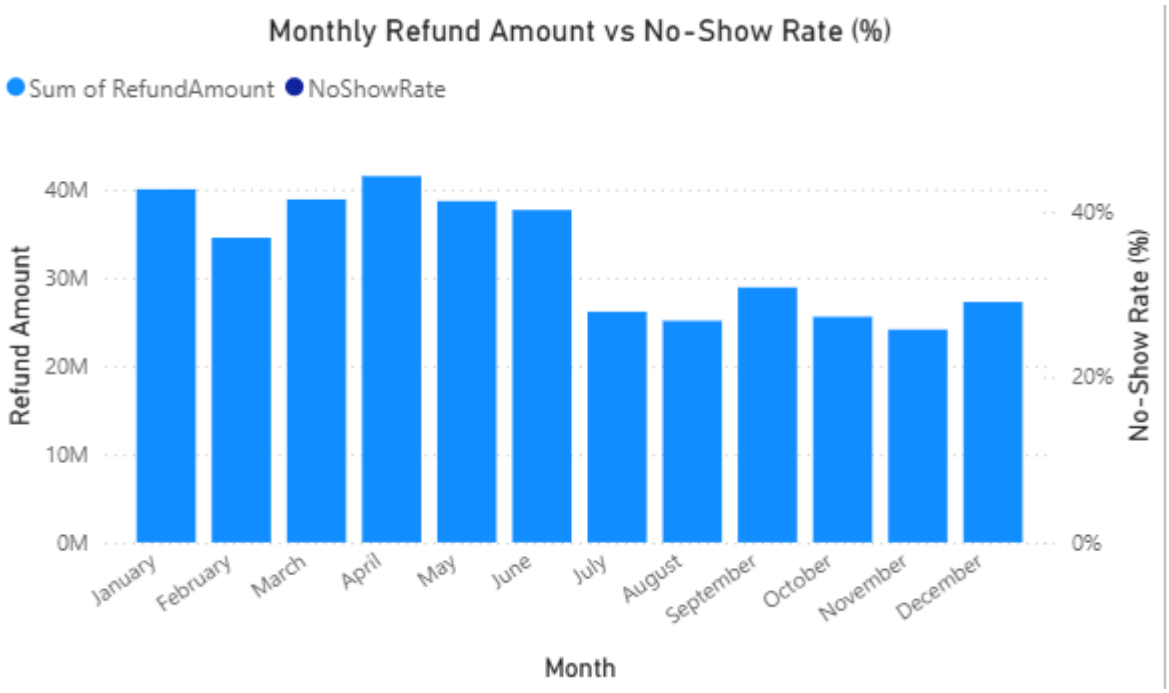
- Some months show high refunds → financial leakage
- No-show rate varies but remains significant
- Weak correlation between refunds and no-shows

Problem Identified:

- Revenue loss due to cancellations and no-shows
- Rooms remain unused despite demand

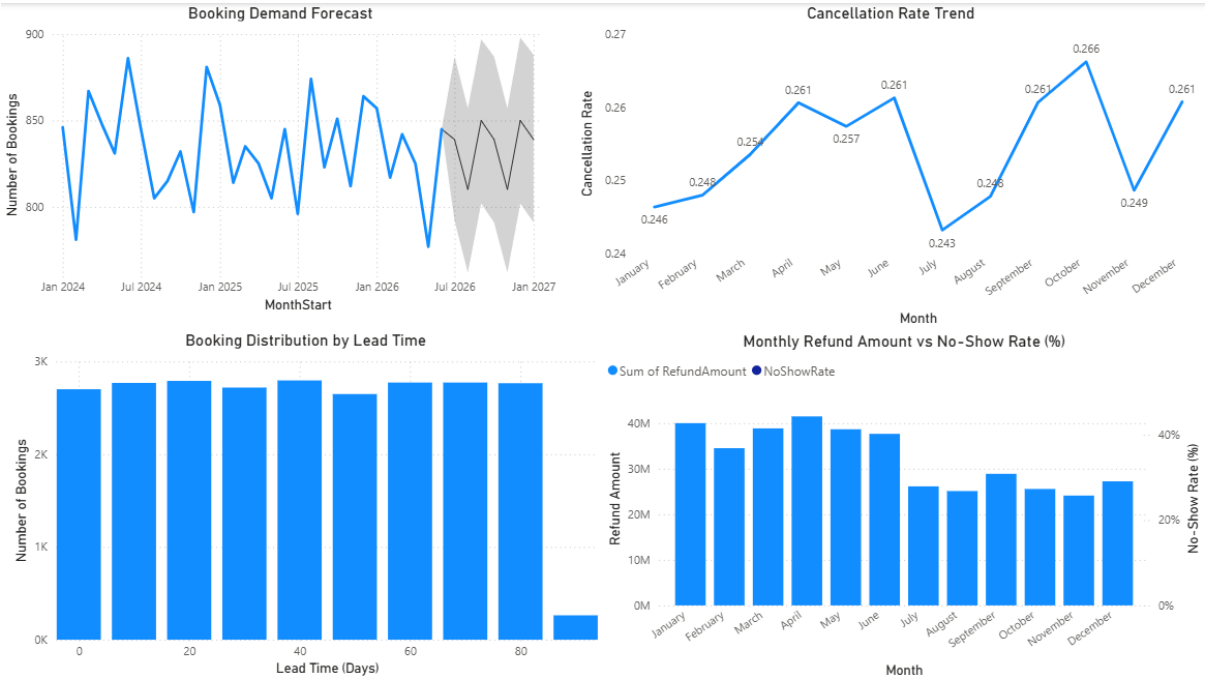
Solution:

- Introduce partial refund policies
- Overbooking strategy (controlled risk)
- Use predictive models to identify high-risk customers



5. Key Insights

- Booking demand is stable but lacks strong growth
- Cancellation rates fluctuate seasonally
- Lead time analysis shows opportunity for better pricing strategies
- Refunds and no-shows significantly impact revenue
- Combining financial and operational metrics gives better decision-making insight.



6. Conclusion

This analysis highlights critical areas affecting hotel performance, including demand forecasting, cancellations, and customer behavior. By leveraging these insights, businesses can optimize pricing, improve customer engagement, and minimize revenue loss.

Implementing strategic solutions such as dynamic pricing, stricter cancellation policies, and predictive analytics can significantly enhance operational efficiency and profitability.